

Generation Interconnection System Impact Study Report

For Queue Project Q18-1017

E Belleville 100 kV

June 18, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1017. The Interconnection Customer proposes a 134.5 MW natural gas combined cycle facility interconnecting at the E Belleville 100 kV substation in the Illinois area. The request was received on June 24, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 422 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 15 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,525,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
MONTPELIER ELECTRIC GENERATING STA / MUSKOGEE 500 KV line	LONETREE / SABANERA 500 KV	119.4	3,404	10.2
TRUMAN / LOOKINGGLASS 230 KV line	BEAVER / TREGO 500 KV	111.8	911	9.9
E BELLEVILLE / DARLINGTON COUNTY 100 KV	ALSIP / CRYSTAL VLY 100 KV Line	111.7	119	55.4
TRUMAN / WALDO 230 KV	TREGO / BEAVER 500 KV	110.0	949	9.9
CANAL ROAD / WEEDPATCH 230 KV	BEAVER / TREGO 500 KV L	106.6	1,217	7.6
LARSON STREET SUBSTATION / NORWICH 230 KV	BARRETT SUBSTATION / NORWICH SUBSTATION 500 KV	105.7	1,299	8.9
SWAN VALLEY / MUSKOGEE 500 KV	LONETREE / SABANERA 500 KV Line	105.0	3,576	11.8
TREGO / SWAN VALLEY 500 KV	TROY ENERGY LLC / RICHLAND 500 KV line	104.9	4,453	19.4
OMNI-HACKBERRY / NORWICH 500 KV	TREGO / BEAVER 500 KV	103.8	3,369	18.8
MYRTLE PT / GALION GENERATING STATION 230 KV line	BEAVER / TREGO 500 KV	103.3	1,298	7.4
JOSO / EUFAULA SUBSTATION 230 KV line	BEAVER / EUFAULA 500 KV	103.1	783	8.1
IMPERIL / GRUVER 230 KV	IMPERIAL / EUFAULA 500 KV Line	101.4	574	6.6
CORRIDOR / LARSON STREET SUBSTATION 230 KV Line	BEAVER / TREGO 500 KV	101.3	947	9.9
NORWICH 230/1kV Auto Bank 1	BARRETT / NORWICH 500 KV	101.2	995	7.0
LOOKINGGLASS SUBSTATION / CORRIDOR SUBSTATION 230 KV Line	TREGO / BEAVER 500 KV	100.8	980	9.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MONTPELIER ELECTRIC GENERATING STATION / MUSKOGEE 500 KV	Rebuild line with higher-capacity conductor	\$199,303,000	1% (\$1,277,000)
TRUMAN / LOOKNGGLASS 230 KV	Rebuild line with higher-capacity conductor	\$21,455,000	1% (\$194,000)
E BELLEVILLE / DARLINGTON COUNTY 100 KV	Rebuild line with higher-capacity conductor	\$1,580,000	88% (\$1,397,000)
TRUMAN / WALDO 230 KV	Rebuild line with higher-capacity conductor	\$25,798,000	1% (\$233,000)
CANAL ROAD / WEEDPATCH 230 KV	Reconductor line	\$4,790,000	1% (\$42,000)
LARSON STREET / NORWICH 230 KV	Reconductor line	\$5,294,000	1% (\$45,000)
MUSKOGEE / SWAN VLY 500 KV	Reconductor line	\$60,745,000	1% (\$426,000)
SWAN VLY / TREGO 500 KV line	Reconductor line	\$38,470,000	1% (\$307,000)
OMNI-HACKBERRY / NORWICH 500 KV	Reconductor line	\$34,506,000	1% (\$295,000)
MYRTLE PT / GALION GENERATING STATION 230 KV line	Reconductor line	\$8,134,000	1% (\$73,000)
JOSO / EUFAULA SUBSTATION 230 KV line	Reconductor line	\$3,023,000	1% (\$27,000)
IMPERIL / GRUVER 230 KV	Reconductor line	\$3,482,000	1% (\$31,000)
CORRIDOR / LARSON STREET SUBSTATION 230 KV Line	Reconductor line	\$6,823,000	1% (\$62,000)
NORWICH 230/1kV Auto Bank 1	Replace transformer with higher-rated unit	\$6,904,000	1% (\$59,000)
LOOKINGGLASS SUBSTATION / CORRIDOR SUBSTATION 230 KV Line	Reconductor line	\$6,296,000	1% (\$57,000)
	Total allocated to project		\$4,525,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.